



University
of Exeter

TM & NLP (COMM040)

Dr. Rodrigo Wilkens

Intended Learning Outcomes



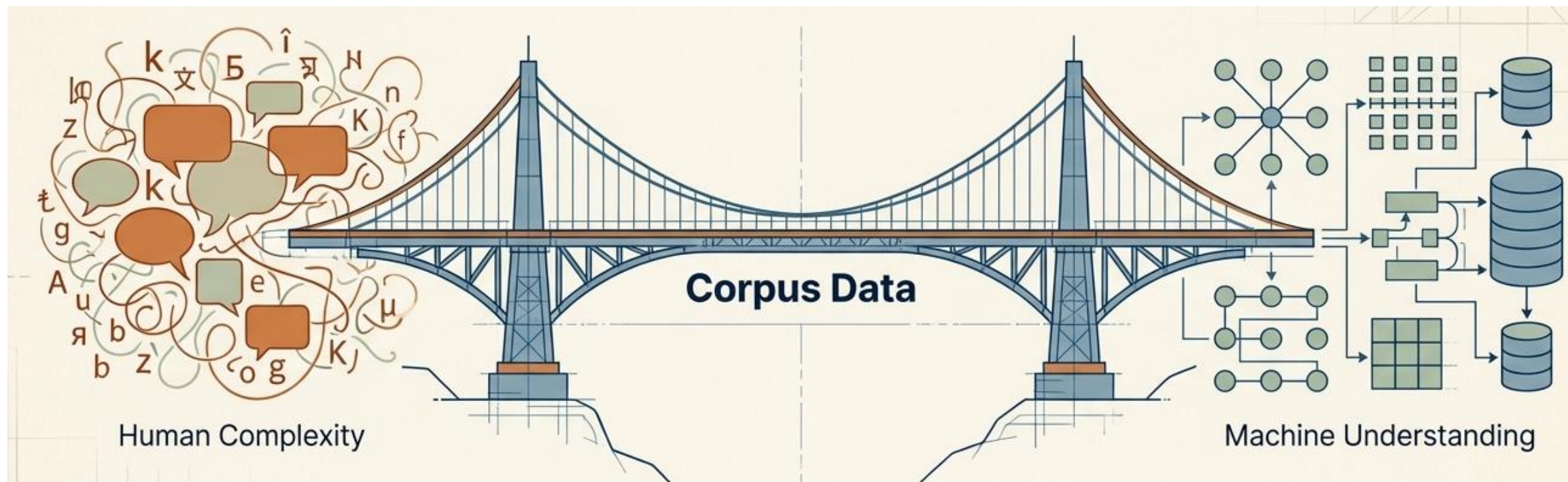
- Understand the strengths, weaknesses and biases of different NLP methods
- Clean, parse, transform and process large collections of text documents
- Know about recent advances and applications of NLP methods, their impact and real world applications.
- Communicate the pros and cons and trade-offs of different approaches to text analysis

Outline

- Corpus
- How to Compile
- Quality Control
- Web as a Corpus
- Dealing with Big Corpora

Outline

- Corpus
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- Corpus data began as a niche linguistic pursuit.
- It is the foundational piece of modern AI.

Definition

- **Corpus:**
 - A collection of machine-readable, authentic texts sampled to be representative of a particular language or variety in a specific context.
- **Corpus in NLP:**
 - A corpus acts as the empirical engine of modern NLP by providing:
 - training data,
 - patterns,
 - contexts...

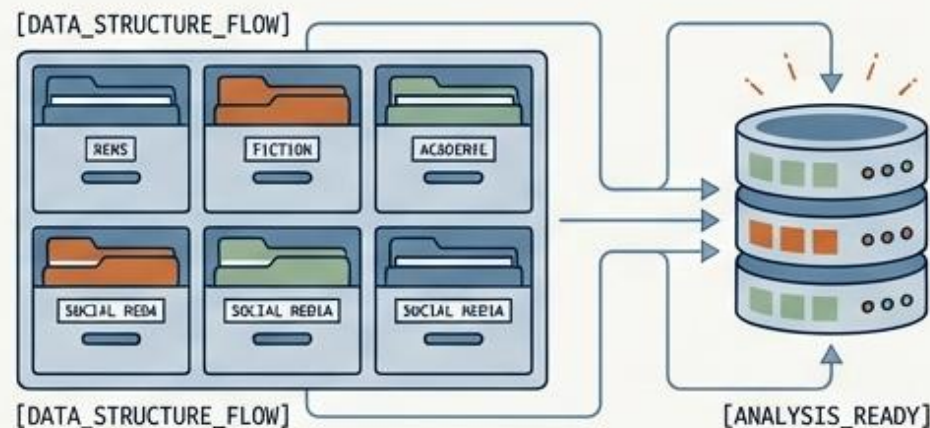
Design parameters

Random Archive



- Unstructured
- Skewed
- Unpredictable

Structured Corpus



- Systematically arranged
- Empirical
- Designed for linguistic analysis

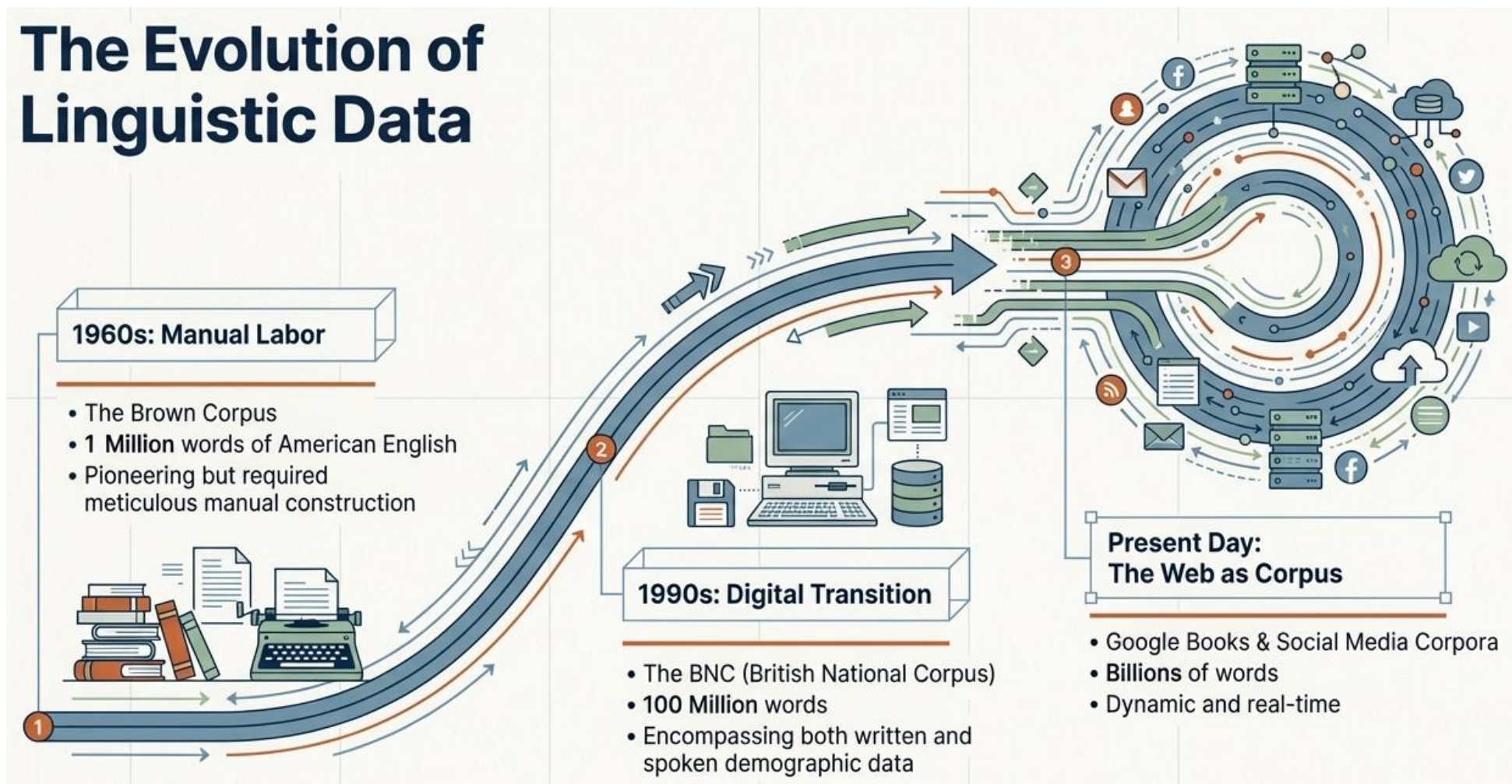
Design parameters

- Bigger is always better (myth)
 - Purpose dictates scale
- A representative corpus captures the full range of variability in a population.
 - Short text chunks (e.g., 2000 words) are sufficient for studying frequent features

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The Evolution of Linguistic Data



Design parameters

Static/Dynamic

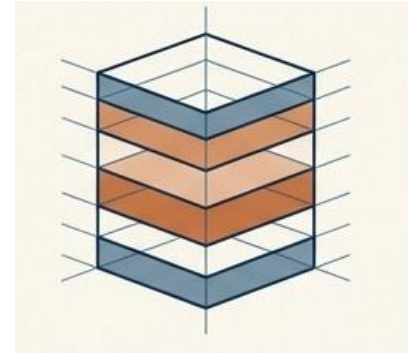
- Sample corpora (static)
 - Fixed snapshot in time
- Monitor corpora (dynamic)
 - Constantly growing

Language content

- General corpora
 - Broad language
- Specialised corpora
 - Domain-specific text (e.g., medical text)

The art of sampling

- Random
 - Warning: it may miss rare important text types
- Stratified random
 - Divided by type of text (e.g., genre)
- Demographic Sampling
 - Sociolects and dialects (e.g., age, sex...)



Text processing

The cats saw the other cat.

Tokens (Total Words)

Every running word and punctuation mark.

Tokens = 7
(The, cats, saw, the, other, cat, .)

Types (Unique Words)

Distinct vocabulary items.
Removes duplicates to measure **lexical richness**.

Types = 5
(The, cats, saw, other, cat.)

Lemmas (Root Words)

The **base dictionary form**.

cats → cat
cat → cat

Lemmas = 4
(the, cat, see, other)

Text processing

Metadata (The Header) [TEI & CES]

Author, publication date,
demographics, and text profile.

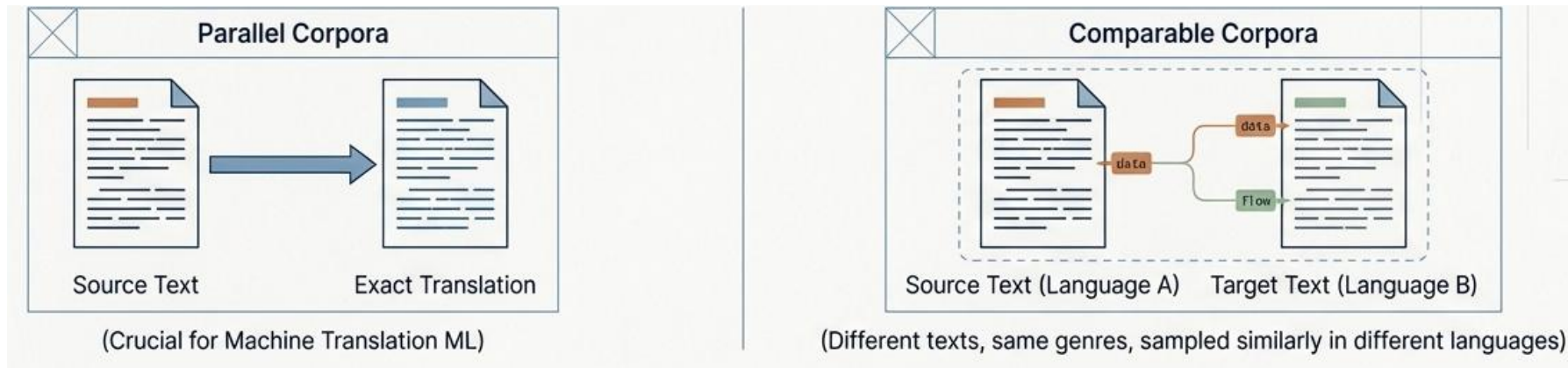
Structural Markup (The Body)

Paragraph boundaries, sentence
breaks, and speaker turns.

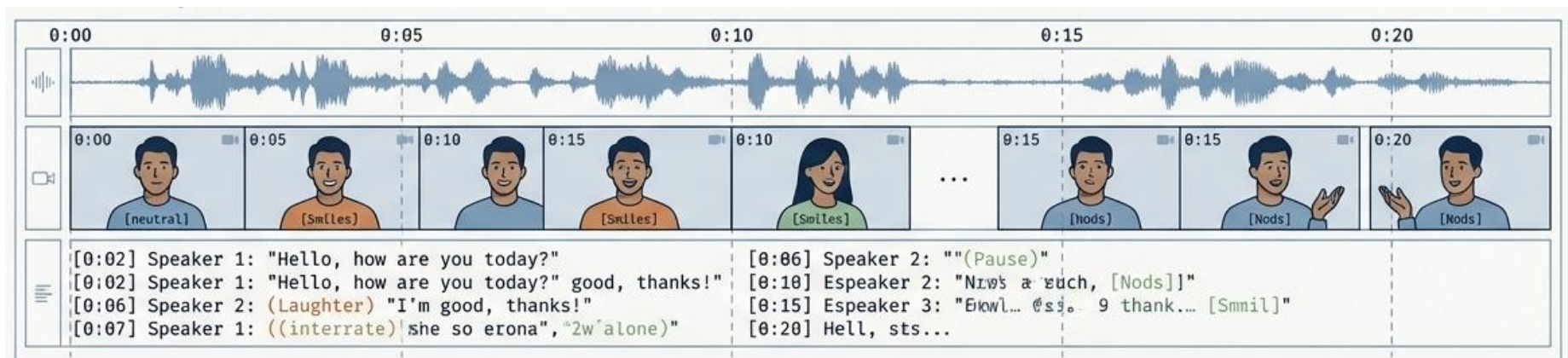
```
<TEI xmlns="http://www.tei-c.org/ns/1.0">
  <teiHeader>
    <fileDesc>
      <titleStmt>
        <title>Noby Dick; or, The Whale</title>
        <author>Herman Melville</author>
      </titleStmt>
      <publicationStmt>
        <publisher>Harper & Brothers</publisher>
        <date when="1851">1851</date>
      </publicationStmt>
      <profileDesc>
        <textClass>
          <catRef target="#novel" scheme="#genre"/>
        </textClass>
        <langUsage>
          <language ident="en">English</language>
        </langUsage>
      </profileDesc>
    </fileDesc>
  </teiHeader>
  <text>
    <body>
      <p>
        <s>Call me Ishmael.</s> <z>Some years ago—never mind how long precisely—having little or no
        money in my purse, and nothing particular to interest me on shore, I thought I would sail
        about a little and see the watery part of the world.</s>
      </p>
      <p>
        <sp>
          <speaker>Captain Ahab</speaker>
          <ab>I'll chase him round Good Hope, and round the Horn, and round the Norway Maelstrom, and
          round perdition's flames before I give him up.</ab>
        </sp>
      </p>
    </body>
  </text>
</TEI>
```

Corpus Alignment

- Multilingual Corpora



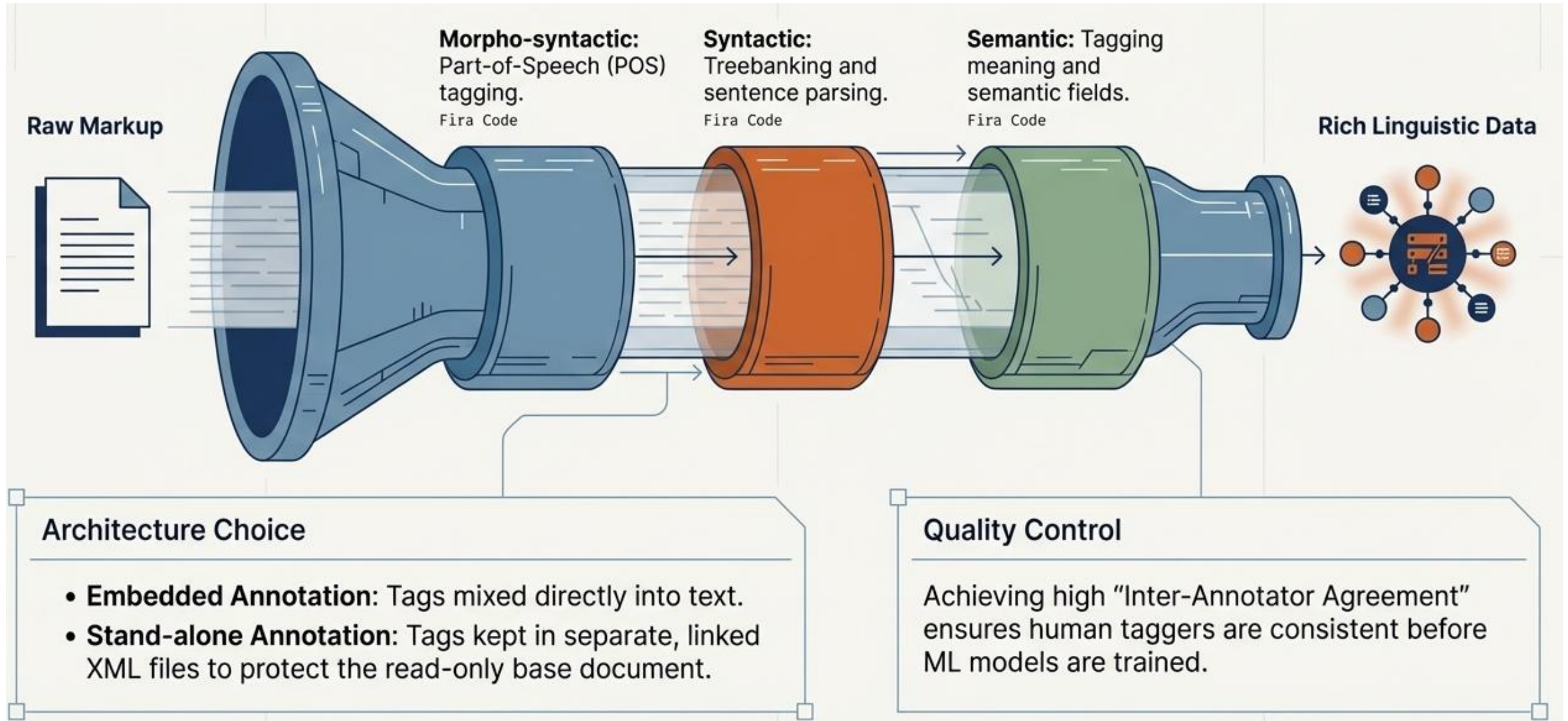
- Multimodal Corpora



Corpus Annotation



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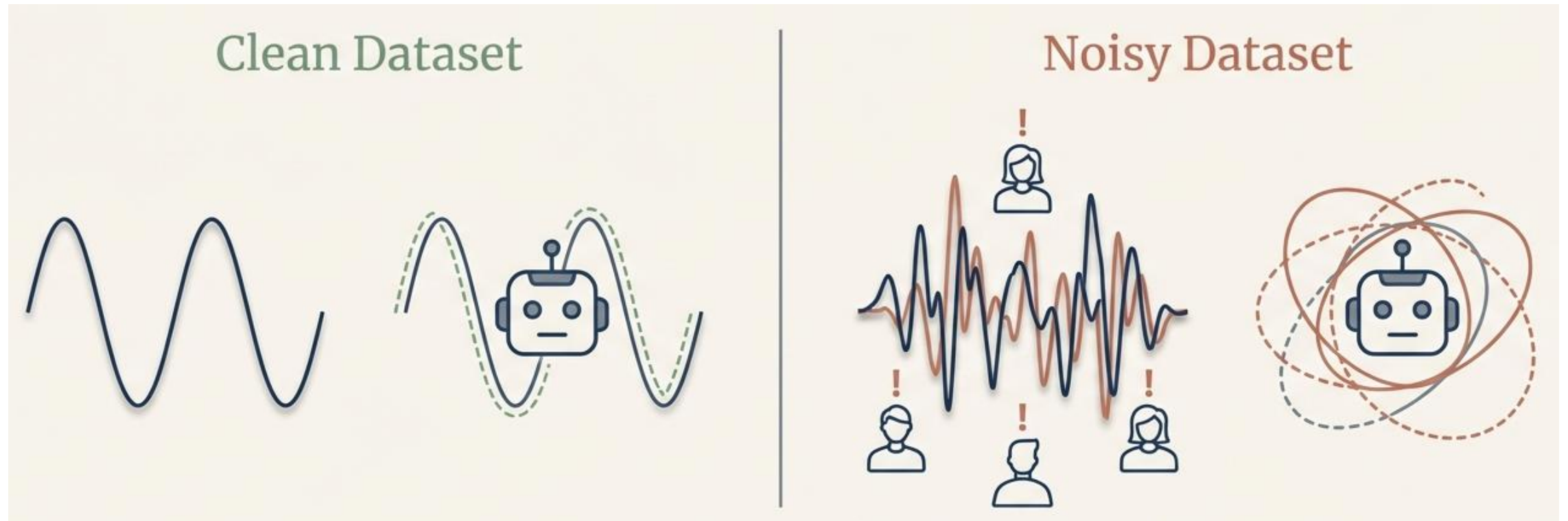


Outline

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Quality Control

- When annotators disagree, datasets become noisy



Quality Control

- When annotators disagree, datasets become noisy
 - High-performing models are falsely penalised for deviating from flawed annotations
 - Leaderboard rankings are unstable

Quality Control

- Inter-Rater Consistency (IRC)



Quality Control

- Causes of disagreement



Subjectivity

Ambiguous tasks like detecting sarcasm or grading emotional severity.



Data Ambiguity

Poorly defined edge cases and complex sentence or image structures.



Task Complexity

Segmenting 3D point clouds vs. binary yes/no tagging.



Annotator Bias

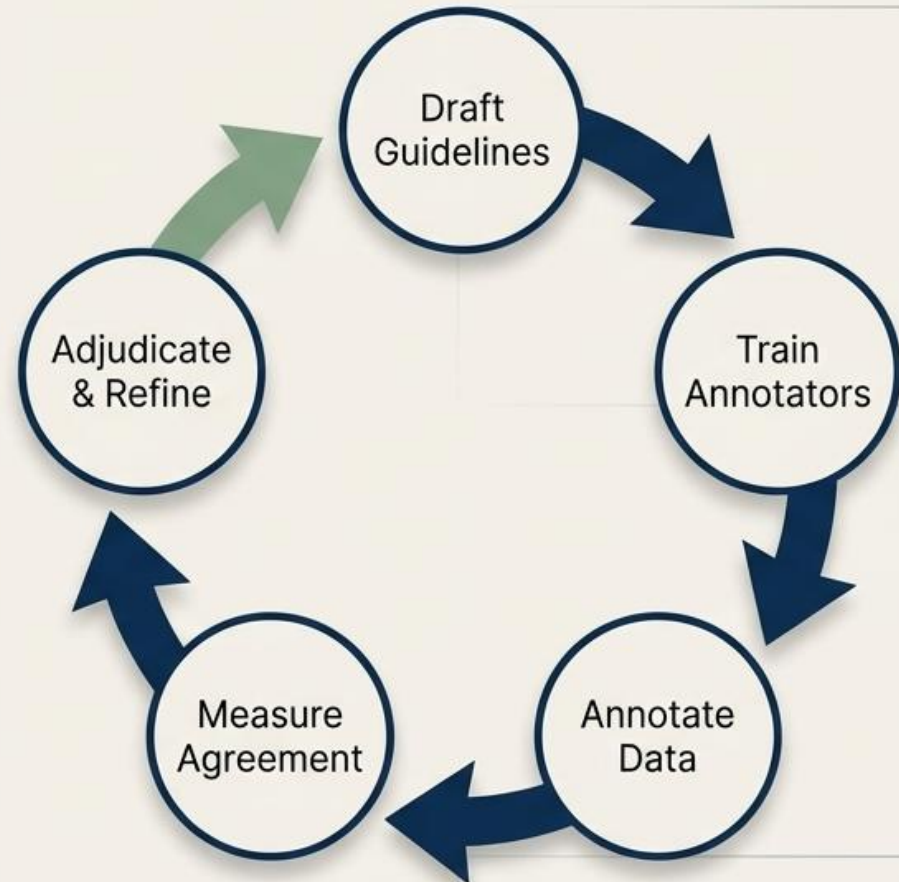
Fatigue, prior knowledge, or drifting interpretation of guidelines.

Quality Control

• Reliable Annotation Process

High agreement is engineered, not accidental. Implement a multi-layered quality assurance process to stabilize human judgment.

- **Comprehensive Protocols:** Draft clear rules, precise definitions, and edge cases.
- **Continuous Calibration:** Run regular alignment sessions and baseline testing.
- **Redundancy & Adjudication:** Assign 3 to 5 raters to complex items and use expert panels to resolve conflicts.



Quality Control



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The Percentage Trap

Simple percent agreement fails to account for agreement by pure chance. In a binary yes/no task, monkeys flipping coins will agree 50% of the time.



Quality Control



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! The Percentage Trap

Simple percent agreement fails to account for agreement by pure chance. In a binary yes/no task, monkeys flipping coins will agree 50% of the time.



! The Correlation Trap

Pearson's r measures linear relationship, not agreement. If Rater B always scores exactly double Rater A, the correlation is a perfect 1.0—despite zero actual agreement.



Quality Control



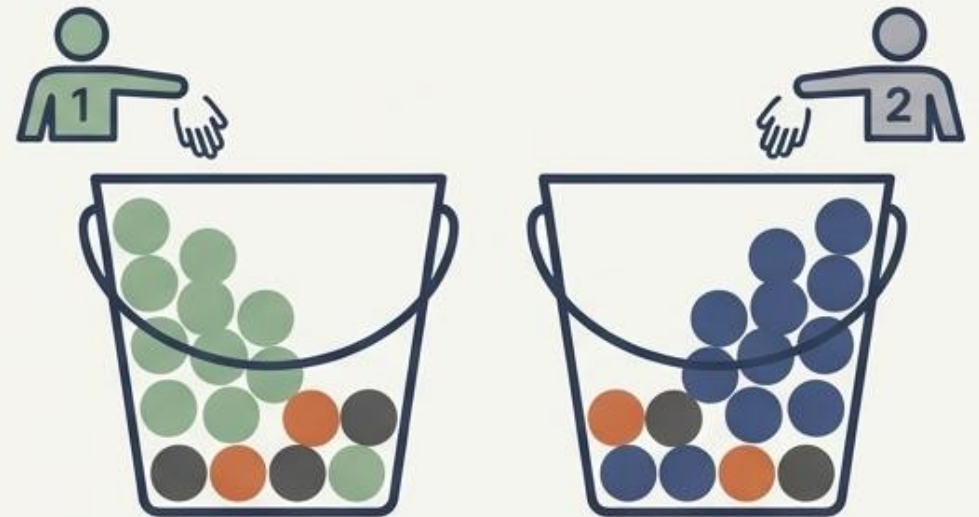
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Arbitrary assignment distinguishes between categories, but not coders.



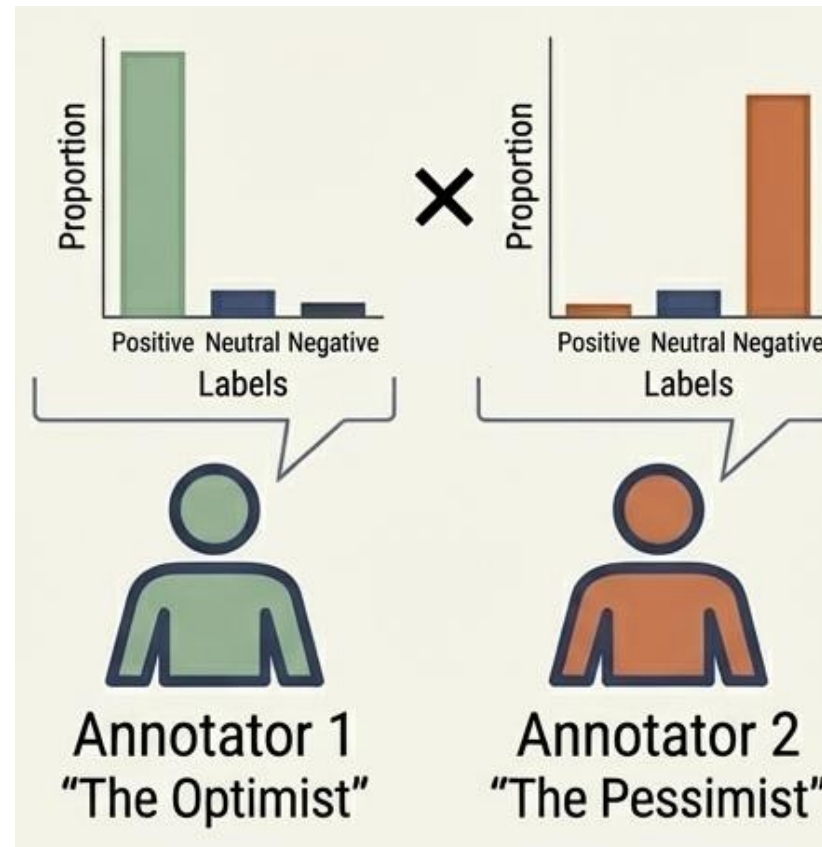
$$P(k|c1) = P(k|c2)$$

Arbitrary assignment is driven by separate prior distributions, accounting for individual coder bias.



$$P(k|c1) \neq P(k|c2)$$

Quality Control



Quality Control



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A_o (Observed Agreement):
The actual proportion of
items on which coders agree.

$$\theta = \frac{A_o - A_e}{1 - A_e}$$

A_e (Expected Agreement):
The agreement expected if
coders made arbitrary or
random choices.

1 - A_e: The total possible
agreement mathematically
achievable beyond chance.



Quality Control

- Cohen's Kappa

$$\kappa \equiv \frac{p_o - p_e}{1 - p_e} = 1 - \frac{1 - p_o}{1 - p_e}, \quad p_e = \frac{1}{N^2} \sum_k n_{k1} n_{k2}$$

$$\kappa = \frac{2 \times (TP \times TN - FN \times FP)}{(TP + FP) \times (FP + TN) + (TP + FN) \times (FN + TN)}$$

Quality Control

- Cohen's Kappa

- Example:

- 50 documents
 - 2 people annotate with HS (yes/no)

A \ B	Yes	No
Yes	20	5
No	10	15

$$p_o = \frac{a + d}{a + b + c + d} = \frac{20 + 15}{50} = 0.7$$

A \ B	Yes	No
Yes	a	b
No	c	d

Quality Control

- Cohen's Kappa

- Example:

- 50 documents
- 2 people annotate with HS (yes/no)

A \ B	Yes	No
Yes	20	5
No	10	15

$$p_e = p_{\text{Yes}} + p_{\text{No}} = 0.3 + 0.2 = 0.5$$

$$p_{\text{Yes}} = \frac{a + b}{a + b + c + d} \cdot \frac{a + c}{a + b + c + d} = 0.5 \times 0.6 = 0.3$$

$$p_{\text{No}} = \frac{c + d}{a + b + c + d} \cdot \frac{b + d}{a + b + c + d} = 0.5 \times 0.4 = 0.2$$

A \ B	Yes	No
Yes	a	b
No	c	d

Quality Control

- Cohen's Kappa

- Example:

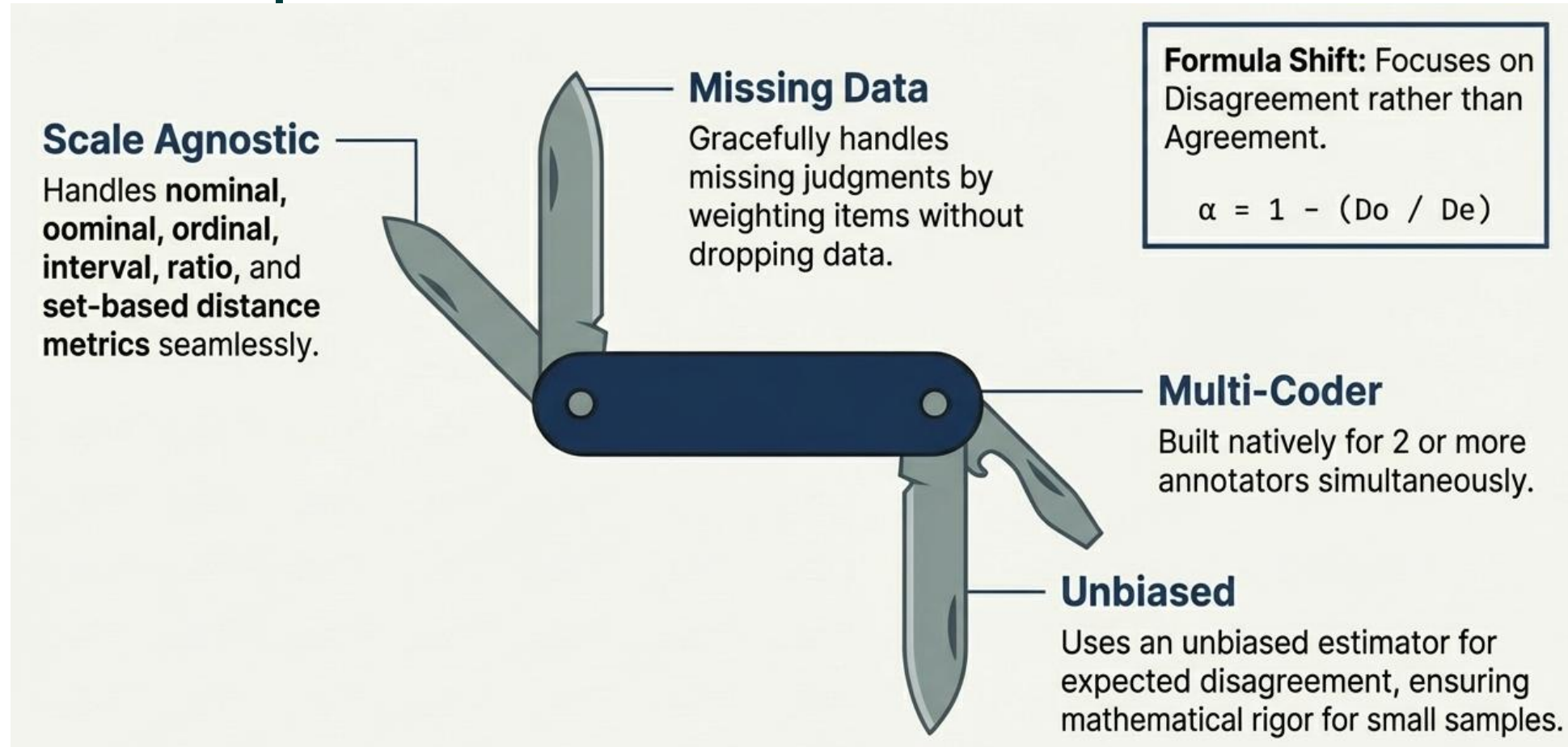
- 50 documents
 - 2 people annotate with HS (yes/no)

A \ B	Yes	No
Yes	20	5
No	10	15

$$\kappa = \frac{p_o - p_e}{1 - p_e} = \frac{0.7 - 0.5}{1 - 0.5} = 0.4$$

Quality Control

- Krippendorff's Alpha

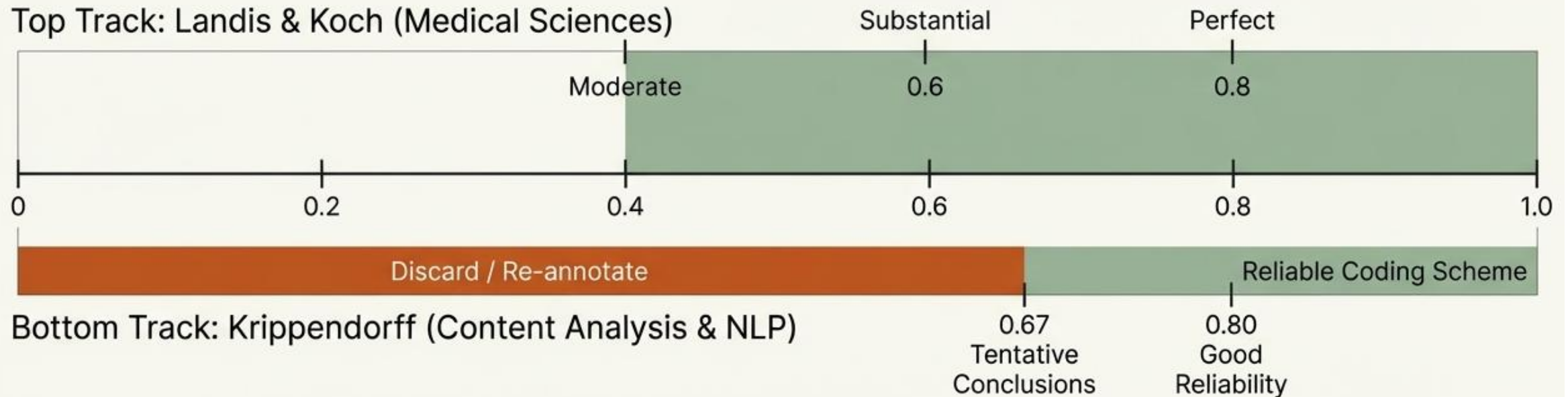


Quality Control

- Krippendorff's Alpha

Interpreting the Magic Numbers

Standard thresholds for coefficient scores.

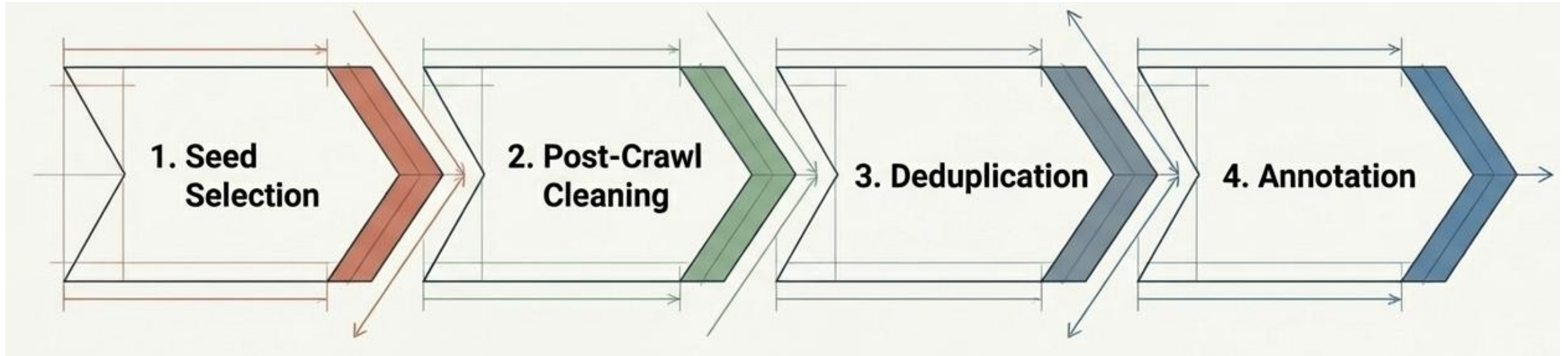


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- How to Compile
- Quality Control
- **Web as a Corpus**
- Dealing with Big Corpora

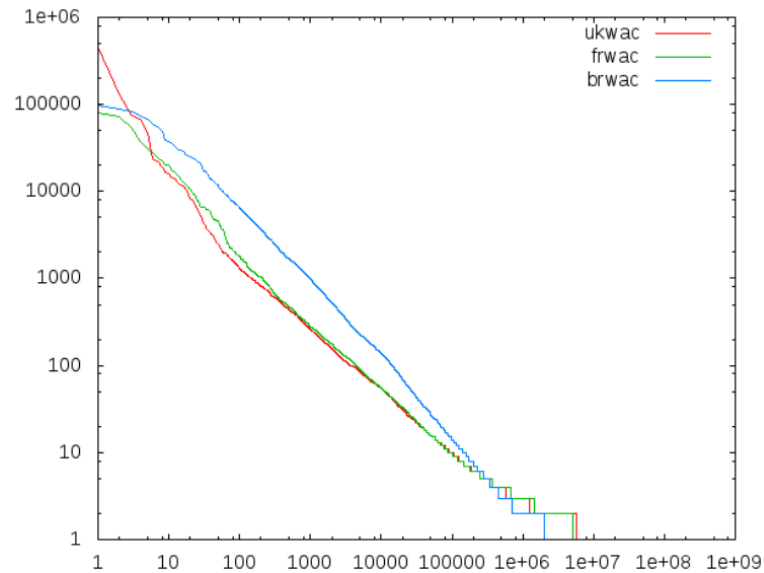
Web as a Corpus

- The WaCky Pipeline

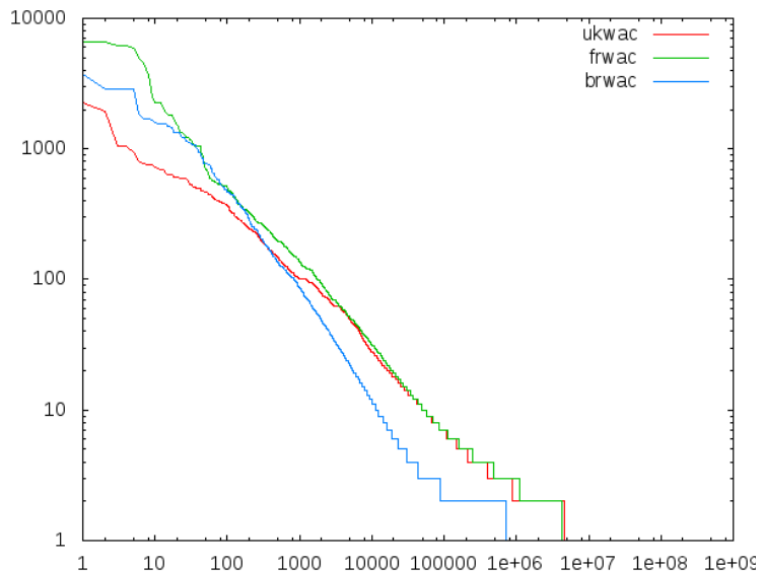


Web as a Corpus

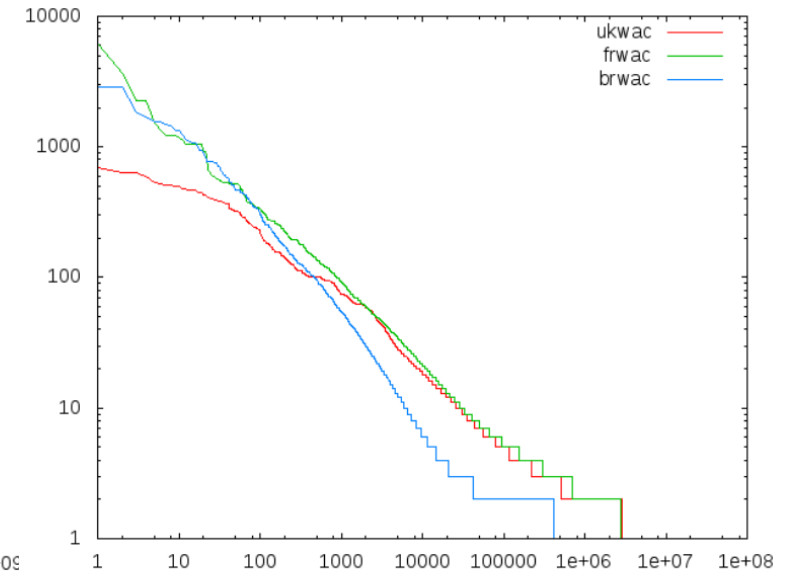
- Deduplication



(a) All sentences



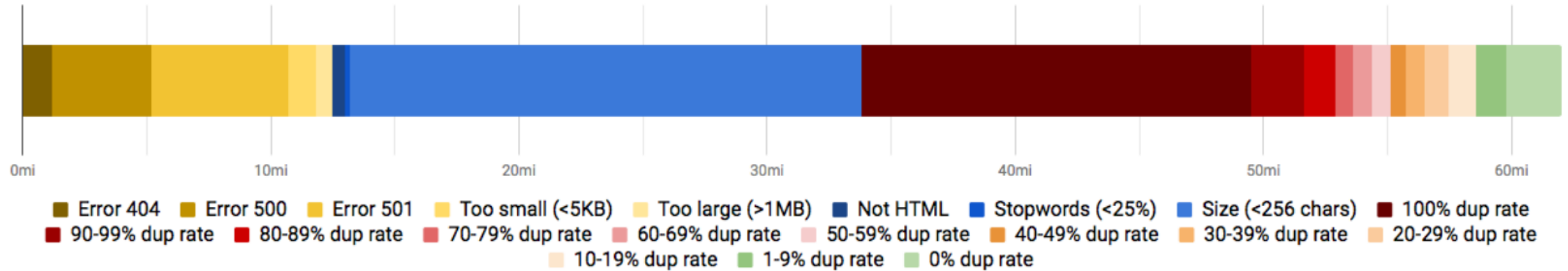
(b) Sentences > 10 tokens



(c) Sentences > 20 tokens

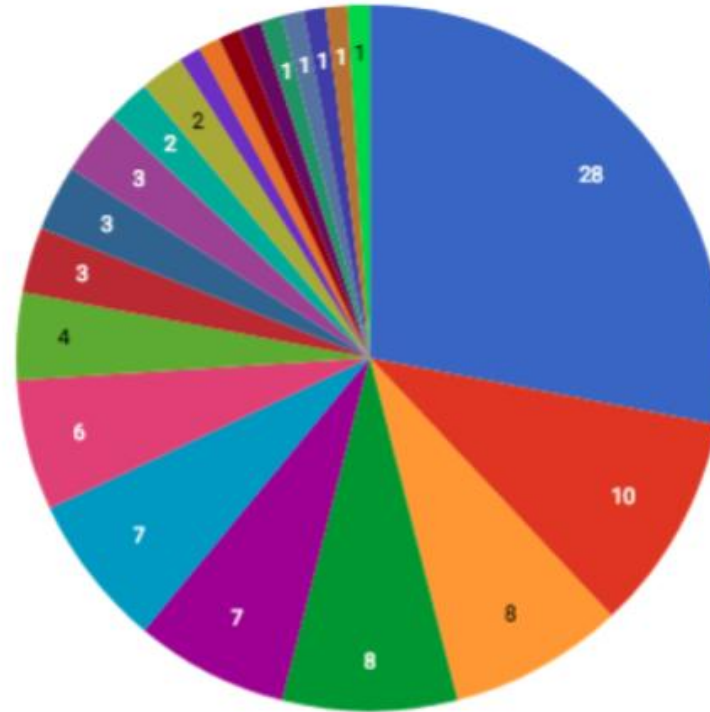
Web as a Corpus

- After long time...



Web as a Corpus

- Random Archive



- News / Weather / Information
- Sports
- Health & Fitness
- Style & Fashion
- Non-Standard Content
- Real Estate
- Arts & Entertainment
- Hobbies & Interests
- Law, Government, & Politics
- Uncategorized
- Careers
- Shopping
- Education
- Technology & Computing
- Illegal Content
- Travel
- Video & Computer Games
- World Football / Soccer
- Personal Finance
- Home & Garden

Web as a Corpus



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- Annotation

```
<text>
  <s id="s1">
    <w id="w1" pos="DT" lemma="the">The</w>
    <w id="w2" pos="JJ" lemma="quick">quick</w>
    <w id="w3" pos="JJ" lemma="brown">brown</w>
    <w id="w4" pos="NN" lemma="fox">fox</w>
    <w id="w5" pos="VBZ" lemma="jump">jumps</w>
    <w id="w6" pos="IN" lemma="over">over</w>
    <w id="w7" pos="DT" lemma="the">the</w>
    <w id="w8" pos="JJ" lemma="lazy">lazy</w>
    <w id="w9" pos="NN" lemma="dog">dog</w>
    <c id="c1">.</c>
  </s>
  <s id="s2">
    <w id="w10" pos="DT" lemma="this">This</w>
    <w id="w11" pos="NN" lemma="sentence">sentence</w>
    <w id="w12" pos="VBZ" lemma="be">is</w>
    <w id="w13" pos="DT" lemma="a">a</w>
    <w id="w14" pos="NN" lemma="pangram">pangram</w>
    <c id="c2"></c>
    <!-- ...more code... -->
  </s>
</text>
```

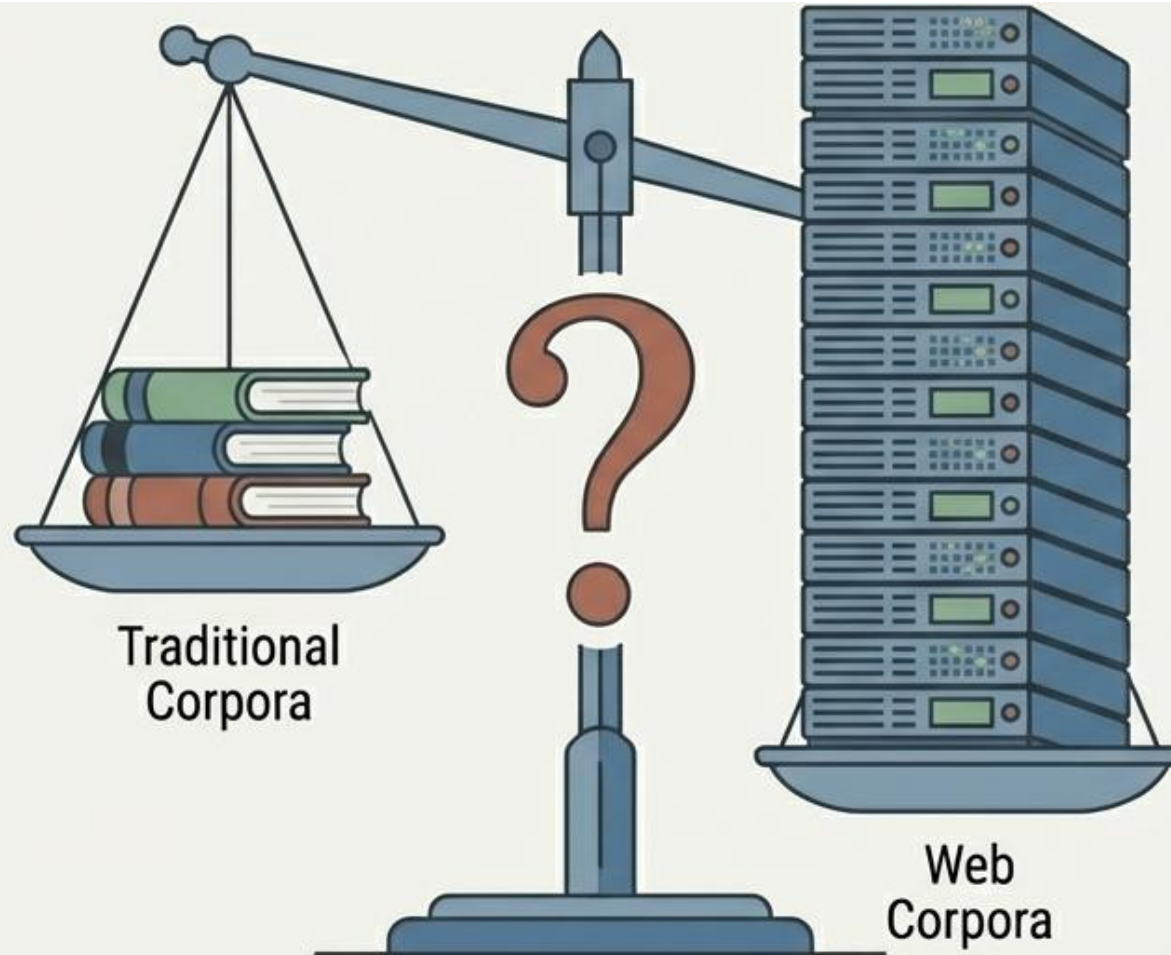
Web as a Corpus



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The Criticism:
“The Web isn’t
representative.”

The Reality: Traditional
corpora are not
representative of
anything other than
themselves, either.

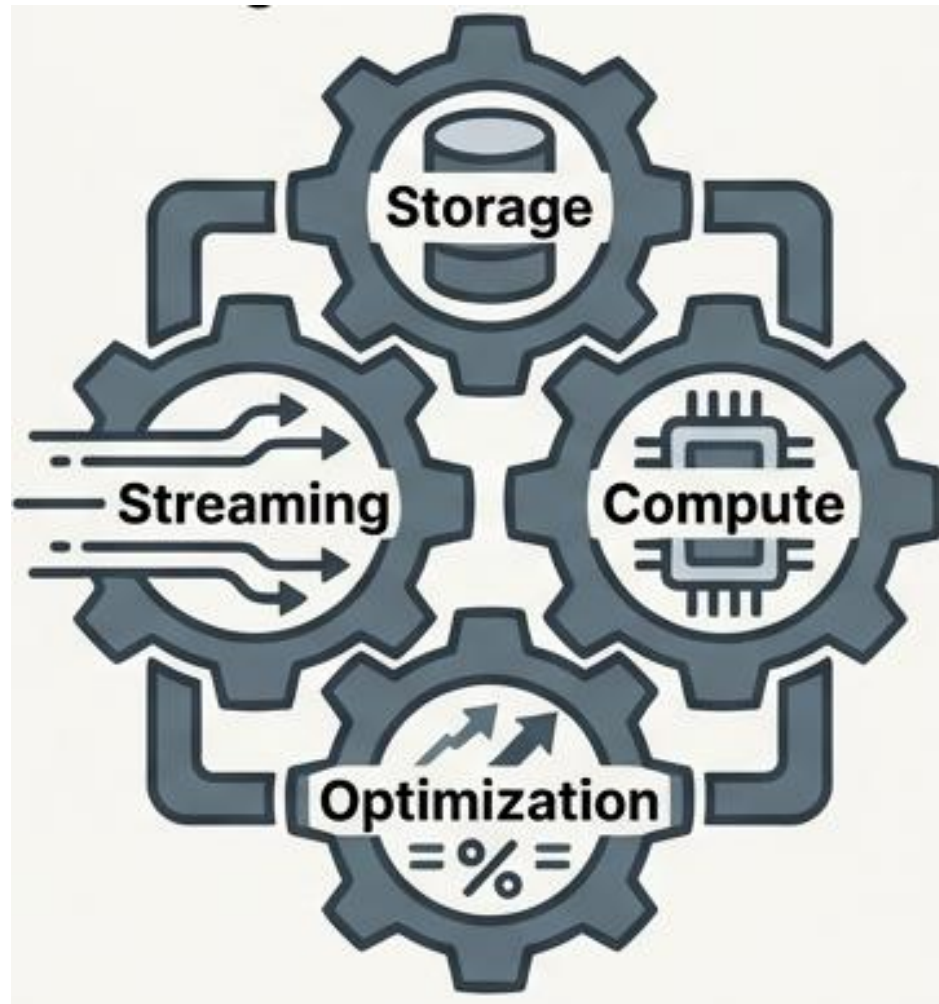


The Paradigm Shift:
Instead of striving for
artificial “balance,”
Web corpora embrace
massive scale and
genre hybridism.
hybridism. This forces
forces the field to
develop mathematical
models for
sublanguages rather
than relying on
arbitrary curation.

Outline

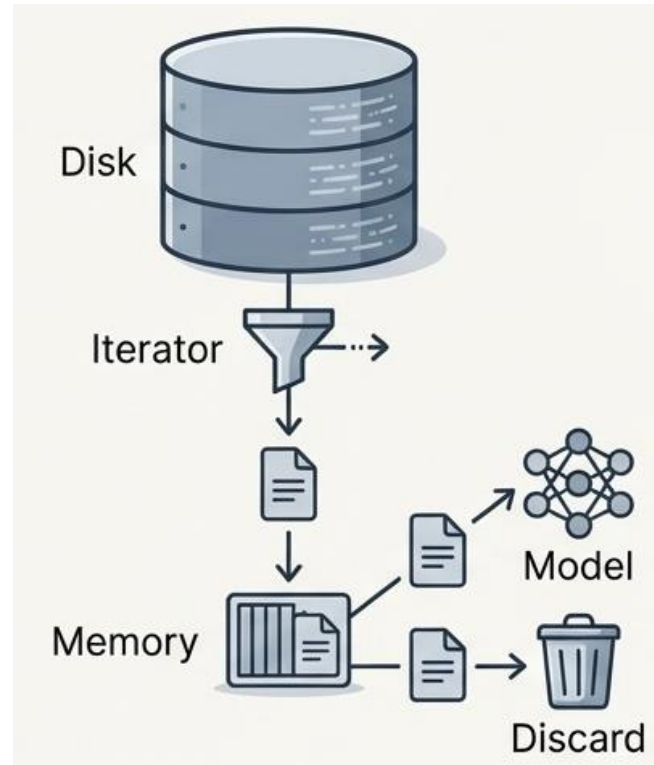
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Dealing with Big Corpora



Dealing with Big Corpora

- Bypassing Memory Limits: Data Streaming

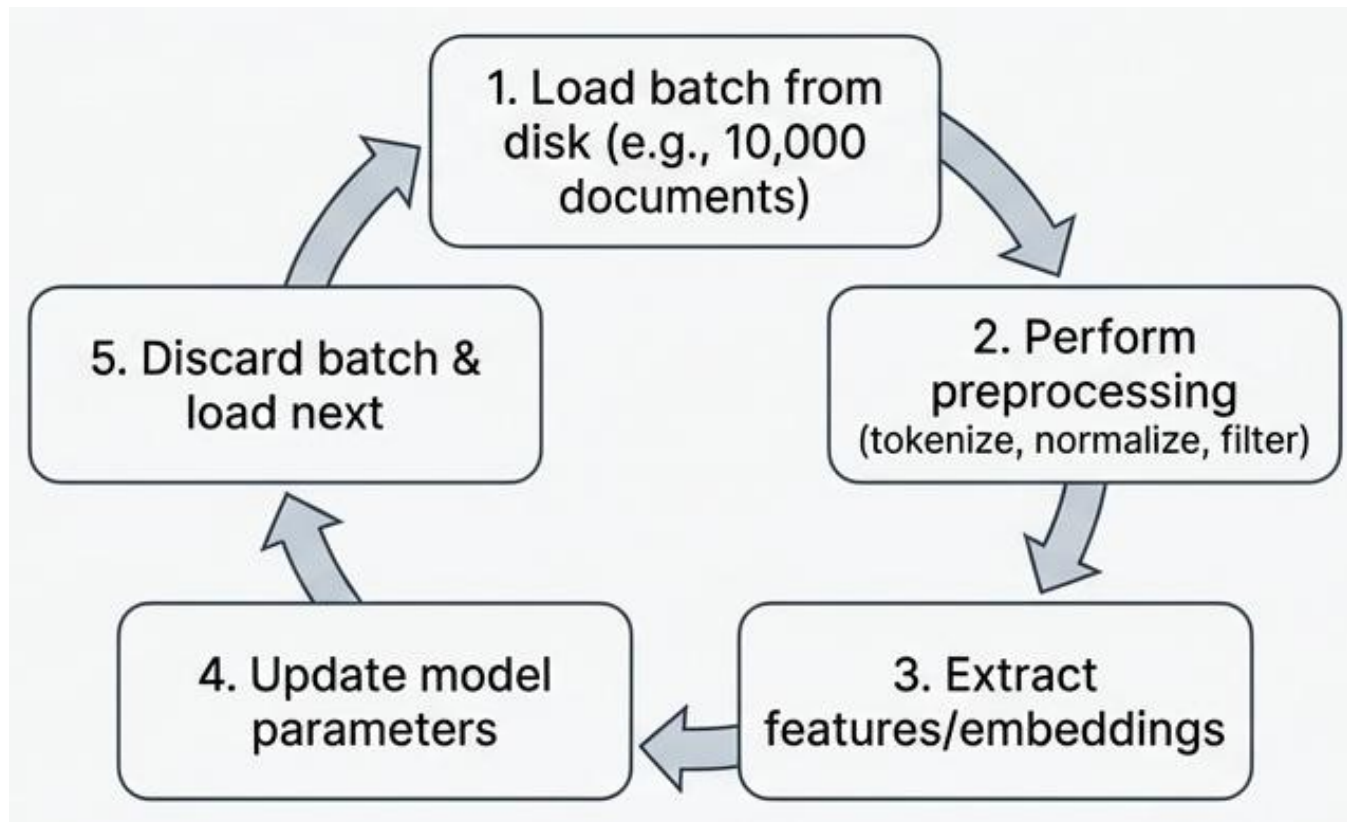


Trade-offs

- **Pros:** Infinite scalability; enables continuous online learning.
- **Cons:** Sequential access makes multi-pass algorithms difficult; heavily bottlenecked by disk I/O speeds.

Dealing with Big Corpora

- Mini-Batch Optimization
 - Low memory usage vs Risk of noisy Gradient



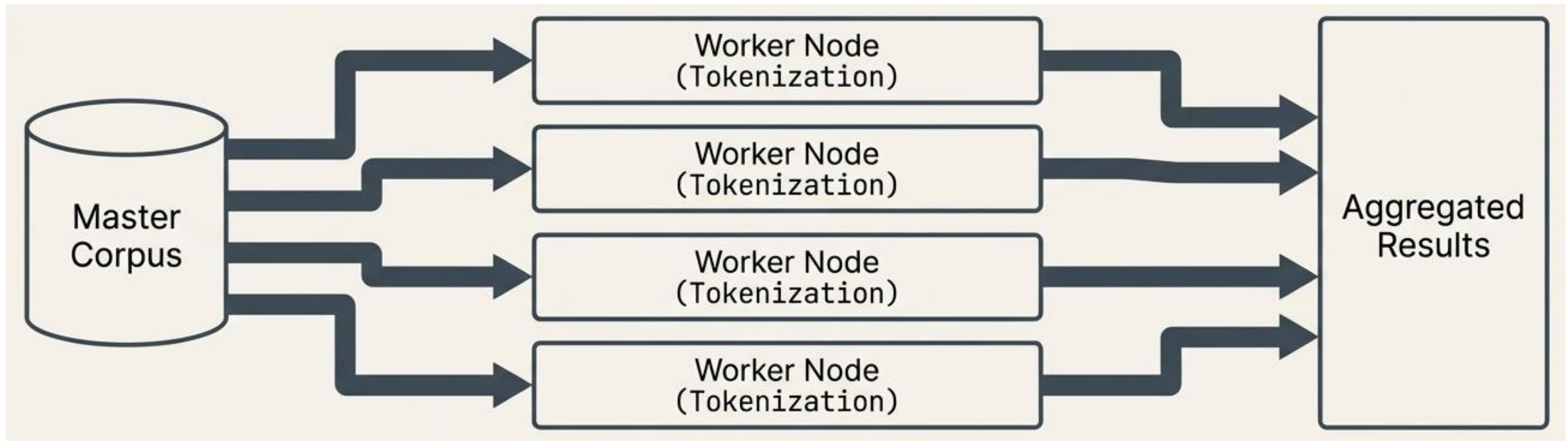
Dealing with Big Corpora

- Optimizing Storage Format

JSON Lines (JSONL)	Parquet	HDF5
Structure: One JSON object per line. Strength: Easy streaming and incremental parsing. Ideal for web-crawled datasets. <pre>{"text": "Natural language..."}</pre>	Structure: Columnar storage. Strength: High compression and selective column access. Perfect for analytical workloads.	Structure: Hierarchical storage. Strength: Partial data loading. Industry standard for precomputed embeddings and massive numerical arrays.

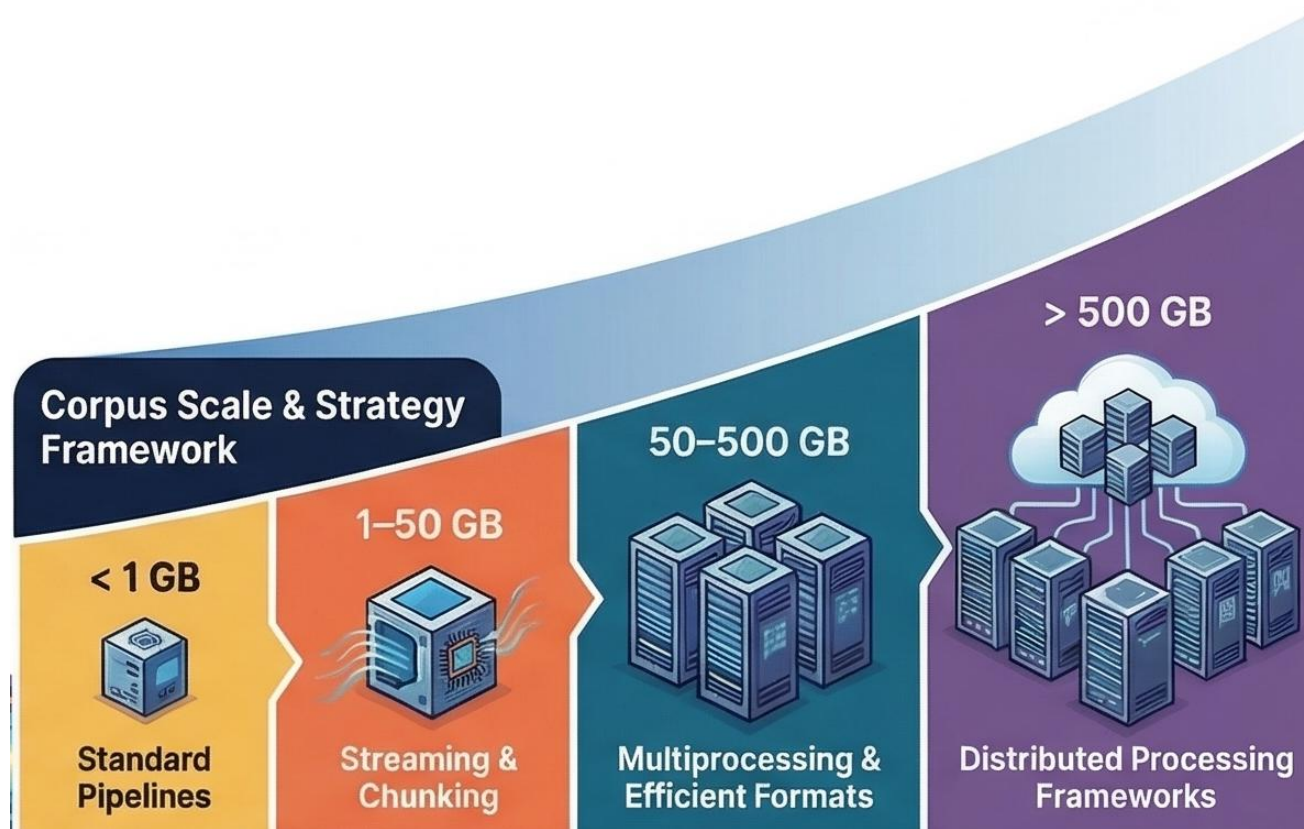
Dealing with Big Corpora

- Distributed Processing



Trade-off: Extreme compute power vs. Network bottlenecks caused by data shuffling and synchronization overhead between nodes.

Dealing with Big Corpora



Conclusion

- By the end of this session, students will be able to:
 - Analyse how corpus design, sampling, and annotation choices introduce bias and affect model performance.
 - Apply tokenisation, lemmatisation, metadata structuring, and alignment principles in corpus preparation.
 - Explain the role of corpora in training modern NLP systems, including web-scale and multilingual resources.
 - Evaluate annotation quality using inter-rater agreement metrics such as Cohen's Kappa and Krippendorff's Alpha.